

16. Rectifiability, weak linear approximation and tangent measures

In the previous chapter we saw that for $\mathcal{H}^m$ measurable sets with finite $\mathcal{H}^m$ measure rectifiability is equivalent to the m -linear approximation property. One of the main themes of this chapter is to prove that the weak m -linear approximation property (recall Definition 15.10) is enough to imply rectifiability. At the same time we shall derive information about densities and projections of rectifiable sets. After that we shall essentially reformulate these approximation properties in terms of tangent measures. Below m and n will be integers with $0 < m < n$.

A lemma on projections of purely unrectifiable sets

Note that after we have proved Theorem 16.2 the following lemma will become worthless: only sets with zero $\mathcal{H}^m$ measure will satisfy its assumptions.

16.1. Lemma. *Let A be an $\mathcal{H}^m$ measurable subset of $\mathbf{R}^n$ with $\mathcal{H}^m(A) < \infty$. If A is purely m -unrectifiable and weakly m -linearly approximable, then $\mathcal{H}^m(P_V A) = 0$ for all $V \in G(n, m)$.*

Proof. Let $0 < \varepsilon < 1/2$ and $V \in G(n, m)$. Using the weak m -linear approximation property 15.10 we can find a compact subset C of A and positive numbers r_0 , δ and η with $\eta < \delta\varepsilon < \varepsilon$, $\mathcal{H}^m(A \setminus C) < \varepsilon$, for which the following holds: if $a \in C$ and $0 < r < r_0$, then

$$(1) \quad \mathcal{H}^m(A \cap B(a, r)) > \delta r^m$$

and there is $W \in A(n, m)$ such that $a \in W$ and

$$(2) \quad C \cap B(a, r) \setminus W(\eta r) = \emptyset.$$

In fact, if one chooses C , r_0 and δ so that for $a \in C$ and $0 < r < r_0$ (1) holds (using only the positiveness of lower density for A) and, by (15.9), that

$$\mathcal{H}^m(A \cap B(a, 2r) \setminus W(\eta r/2)) < \delta(\eta r/2)^m,$$

then $B(b, \eta r/2) \subset B(a, 2r) \setminus W(\eta r/2)$ for $b \in B(a, r) \setminus W(\eta r)$ and (2) follows from (1). Then we also have

$$(3) \quad \mathcal{H}^m(P_V(A \setminus C)) < \varepsilon.$$

Since C is purely m -unrectifiable, it follows from Lemma 15.13 that

$$\mathcal{H}^m\left(\bigcup_{i=1}^{\infty}\{a \in C : C \cap X(a, 1/i, V^\perp, \eta) = \emptyset\}\right) = 0.$$

Hence for $\mathcal{H}^m$ almost all $a \in C$ there are points $b \in C$ arbitrarily close to a such that

$$(4) \quad |P_V(b - a)| < \eta |b - a|.$$

Suppose $a, b \in C$ satisfy (4) and $r = |a - b| < r_0$. Let W be as in (2) with $a \in W$ and let $c = P_W b$. Using (2) we see that

$$|c - b| \leq \eta r, \quad r/2 \leq |c - a| \leq r, \quad |P_V(c - a)| < 2\eta r.$$

We can select an orthonormal basis $\{e_1, \dots, e_m\}$ for $W - a$ so that $P_V(e_i) \cdot P_V(e_j) = 0$ for $i \neq j$, see e.g. Federer [3, 1.7.3]. Then for some i ,

$$|P_V e_i| \leq 2r^{-1} |P_V(c - a)| < 4\eta,$$

because otherwise we should have

$$\begin{aligned} |P_V(c - a)|^2 &= \sum_{j=1}^m |(c - a) \cdot e_j|^2 |P_V e_j|^2 \\ &> 4r^{-2} |P_V(c - a)|^2 |c - a|^2 \geq |P_V(c - a)|^2. \end{aligned}$$

It follows that $P_V(W \cap B(a, r))$ is contained in an m -dimensional rectangle with one side of length $8\eta r$ and the others of length $2r$. Hence by condition (2), $P_V(C \cap B(a, r))$ is contained in a rectangle with side-lengths $10\eta r, 2r + 2\eta r, \dots, 2r + 2\eta r$. Therefore, as $\eta < 1/2$ and $\mathcal{H}^m = 2^m \alpha(m)^{-1} \mathcal{L}^m$ on $\mathbf{R}^m$, we have with $c = 20^m \alpha(m)^{-1}$,

$$(5) \quad \mathcal{H}^m(P_V(C \cap B(a, r))) \leq c\eta r^m.$$

We can now use the covering theorem 2.8 to find disjoint balls $B(a_i, r_i)$ satisfying (5) and such that $a_i \in C$ and

$$\mathcal{H}^m\left(C \setminus \bigcup_{i=1}^{\infty} B(a_i, r_i)\right) = 0.$$

Using (5) and (1), we thus obtain

$$\begin{aligned} \mathcal{H}^m(P_V(C)) &\leq \sum_{i=1}^{\infty} \mathcal{H}^m(P_V(C \cap B(a_i, r_i))) \\ &\leq c\eta \sum_{i=1}^{\infty} r_i^m \leq c\eta \delta^{-1} \sum_{i=1}^{\infty} \mathcal{H}^m(A \cap B(a_i, r_i)) \\ &\leq c\varepsilon \mathcal{H}^m(A). \end{aligned}$$

Combining with (3), $\mathcal{H}^m(P_V(A)) < (1 + c\mathcal{H}^m(A))\varepsilon$, and the lemma follows. $\square$

Weak linear approximation, densities and projections

The first part of the following theorem will be essential in Chapter 17.

16.2. Theorem. *Let E be an $\mathcal{H}^m$ measurable subset of $\mathbf{R}^n$ with $0 < \mathcal{H}^m(E) < \infty$. Then E is m -rectifiable if and only if E is weakly m -linearly approximable. Moreover, in this case,*

- (1) $\Theta^m(E, x) = 1$ for $\mathcal{H}^m$ almost all $x \in E$, and
- (2) $\mathcal{H}^m(P_V E) > 0$ for $\gamma_{n,m}$ almost all $V \in G(n, m)$.

We first sketch the basic ideas for the proof that the weak m -linear approximation property implies rectifiability, which is the main part, and a refinement of the argument also gives the rest. In view of Lemma 16.1 it suffices to show that some projection of E has positive $\mathcal{H}^m$ measure. Let F be a compact subset of E where $0 < cr^m \leq \mathcal{H}^m(E \cap B(a, r)) \leq dr^m < \infty$ for $0 < r < r_0$ and where the approximation conditions (15.8) and (15.9) hold uniformly. Consider a ball $B(a, r)$ such that $\mathcal{H}^m((E \setminus F) \cap B(a, r))/r^m$ is small and most of $E \cap B(a, r)$ lies close to an m -plane W . Suppose that the projection of $E \cap B(a, r)$ on W is small. This implies that we can find many disjoint open cylinders C_i of radii $\varrho_i \ll r$ and orthogonal to W such that the cylinders with the same centres and radii $5\varrho_i$ are disjoint, that $F \cap B(a, r) \cap C_i = \emptyset$ and that $B(a, r) \cap \partial C_i$ contains a point e_i of F , see Figure 16.1. Then for a large number M , $E \cap B(e_i, M\varrho_i)$ is well approximated by an m -plane W_i . For most indices i there is very little of E in $B(a, r) \cap C_i$, whence W_i must be almost orthogonal to W . This will give us so many disjoint balls $B(x_{i,j}, \varrho_i) \subset B(a, r)$ with $x_{i,j} \in F$ that $\mathcal{H}^m(E \cap \bigcup_{i,j} B(x_{i,j}, \varrho_i))$ will be much greater than r^m , which is a contradiction.

We now give the detailed proof.

Proof. If E is m -rectifiable, E is m -linearly approximable, whence also weakly, by Theorem 15.11. Assume that E is weakly m -linearly approximable. We shall show that then E is m -rectifiable and (1) and (2) hold, which will prove the theorem.

Let $\varepsilon > 0$. Since E has positive lower density $\mathcal{H}^m$ almost everywhere on E , there are a compact subset F of E with $\mathcal{H}^m(E \setminus F) < \varepsilon$ and positive numbers δ and r_0 such that

- (3) $\mathcal{H}^m(E \cap B(a, r)) > \delta r^m$ for $a \in F$, $0 < r < r_0$.

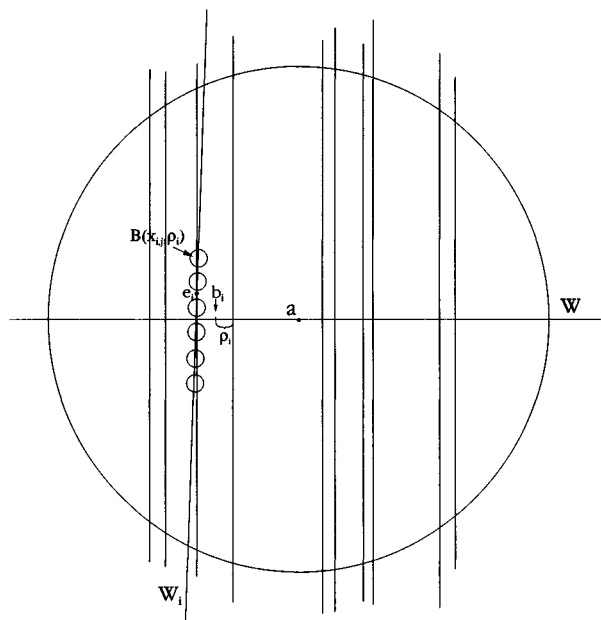


Figure 16.1.

Let $\eta > 0$, $1/2 < u < 1$, $0 < \gamma \leq 1$, with $\eta < \gamma(1 - u)/8$. By Theorem 6.2 (2), as in the proof of Lemma 16.1, we can find a compact subset F_1 of F and a positive number $r_1 \leq r_0$ such that $\mathcal{H}^m(F \setminus F_1) < \varepsilon$ and that for any $a \in F_1$, $0 < r < r_1$, there is $W \in A(n, m)$ for which $a \in W$ and

$$(4) \quad F_1 \cap B(a, 2r) \setminus W(\eta r) = \emptyset,$$

$$(5) \quad W \cap B(a, r) \subset F(\eta r).$$

Note that when η is fixed, $\lambda = \lambda(a)$ in Definition 15.10 depends only on a , and we can first take a large subset F' of F such that $\lambda(a) \geq \lambda_0 > 0$ for $a \in F'$ and then apply $\Theta^m(E \setminus F, x) = 0$ for $\mathcal{H}^m$ almost all $x \in F$ to find $F_1 \subset F'$ so that (5) holds for F . By Theorem 6.2 for $\mathcal{H}^m$ almost all $a \in F_1$, $\Theta^{*m}(E, a) \leq 1$ and $\Theta^m(E \setminus F_1, a) = 0$. Hence, as before, we see that for $\mathcal{H}^m$ almost all $a \in F_1$ there exists a positive number $r_2 \leq r_1$ such that for $0 < r < r_2$ there is an affine subspace $W \in A(n, m)$ with $a \in W$ for which

$$(6) \quad F_1 \cap B(a, r) \setminus W(\eta r) = \emptyset,$$

$$(7) \quad W \cap B(a, r) \subset F_1(\eta r),$$

$$(8) \quad \mathcal{H}^m(E \cap B(a, r)) < 3^m r^m,$$

$$(9) \quad \mathcal{H}^m((E \setminus F_1) \cap B(a, r)) < 400^{-m} t \delta r^m,$$

with $t = 2^m \gamma^m (u^m - u^{2m})$.

Fix such a , r and W , and let $V \in G(n, m)$ be such that $P_V|W: W \rightarrow V$ is one-to-one and $\gamma \leq \|(P_V|(W - a))^{-1}\|^{-1}$. Then

$$(10) \quad |P_V x - P_V y| \geq \gamma |x - y| \quad \text{for } x, y \in W.$$

We shall show that if for given δ , u and γ , the number η is sufficiently small, then

$$(11) \quad \mathcal{H}^m(P_V(E \cap B(a, r))) \geq (2\gamma u^2 r)^m.$$

This will prove all the assertions. First, the rectifiability follows from Lemma 16.1; if E were not m -rectifiable we could apply (11) to a purely m -unrectifiable subset of E with positive $\mathcal{H}^m$ measure. Secondly, choosing V so that $V + a = W$, we can take $\gamma = 1$, which gives for $0 < r < r_2$,

$$\mathcal{H}^m(E \cap B(a, r)) \geq (2u^2 r)^m$$

and leads to (1) as $\Theta^{*m}(E, x) \leq 1 \mathcal{H}^m$ almost everywhere. Finally (2) follows since $P_V|W$ is one-to-one for $\gamma_{n,m}$ almost all $V \in G(n, m)$ by Corollary 3.14, whence $\gamma_{n,m}(\{V : \|(P_V|(W - a))^{-1}\|^{-1} < \gamma\}) \rightarrow 0$ as $\gamma \downarrow 0$, and this measure depends only on γ .

Suppose that (11) fails. Set

$$C = P_V(F_1 \cap B(a, r)) \quad \text{and} \quad D = P_V(W \cap B(a, ur)) \setminus C.$$

Then C is compact and

$$\mathcal{H}^m(C) < (2\gamma u^2 r)^m.$$

By (10), $V \cap B(P_V a, \gamma ur) \subset P_V(W \cap B(a, ur))$, whence, recalling that $t = 2^m \gamma^m (u^m - u^{2m})$,

$$(12) \quad \mathcal{H}^m(D) \geq tr^m.$$

(This is the only place in the book where we really need the precise value of the constant in 4.3 (2).) We can cover D with balls $B(b, \varrho)$ such that $b \in D$, $C \cap U(b, \varrho) = \emptyset$ and $C \cap \partial B(b, \varrho) \neq \emptyset$. Applying the covering theorem 2.1 to the balls $B(b, 5\varrho)$, we can select a finite number of them, say $B(b_i, 5\varrho_i)$, $i = 1, \dots, p$, such that

$$(13) \quad B(b_i, 5\varrho_i) \cap B(b_j, 5\varrho_j) = \emptyset \quad \text{for } i \neq j,$$

and the balls $B(b_i, 25\rho_i)$ cover D , whence by (12)

$$(14) \quad \sum_{i=1}^p \varrho_i^m \geq 50^{-m} t r^m.$$

From (7) and the inequality $\eta < 1 - u$, we obtain

$$(15) \quad \varrho_i \leq \eta r \quad \text{for } i = 1, \dots, p.$$

We now consider the sets

$$S_i = P_V^{-1}(B(b_i, \varrho_i/2)) \cap W(\gamma(1-u)r/4)$$

and rearrange them so that they contain no points of F for $i = 1, \dots, q$ and that they contain at least one point, c_i , of F for $i = q+1, \dots, p$. Letting $b'_i \in W$ be the point with $P_V b'_i = b_i$, we have $b'_i \in B(a, ur)$ and by (10) and (15) for $i = q+1, \dots, p$,

$$\begin{aligned} |a - c_i| &\leq |a - b'_i| + |b'_i - P_W c_i| + |P_W c_i - c_i| \\ &\leq ur + |b_i - P_V(P_W c_i)|/\gamma + (1-u)r/4 \\ &\leq ur + |b_i - P_V c_i|/\gamma + |P_V(c_i - P_W c_i)|/\gamma + (1-u)r/4 \\ &\leq ur + \varrho_i/(2\gamma) + (1-u)r/2 \\ &\leq \eta r/\gamma + (1+u)r/2. \end{aligned}$$

Since $\eta < \gamma(1-u)/3$, this gives

$$B(c_i, \varrho_i/4) \subset B(a, r).$$

Moreover

$$(16) \quad P_V(B(c_i, \varrho_i/4)) \subset V \cap U(b_i, \varrho_i) \subset V \setminus C.$$

It follows that

$$(17) \quad \bigcup_{i=q+1}^p E \cap B(c_i, \varrho_i/4) \subset (E \setminus F_1) \cap B(a, r).$$

Since the balls $B(c_i, \varrho_i/4)$ are disjoint by (16) and (13), we obtain from (3), (17) and (9) that

$$\delta 4^{-m} \sum_{i=q+1}^p \varrho_i^m < 400^{-m} t \delta r^m,$$

whence by (14)

$$(18) \quad \sum_{i=1}^q \varrho_i^m > 100^{-m} t r^m.$$

From now on we only consider $i = 1, \dots, q$. Recalling (6) and how the balls $B(b_i, \varrho_i)$ were chosen, we see that there are points

$$(19) \quad e_i \in P_V^{-1}(\partial B(b_i, \varrho_i)) \cap W(\eta r) \cap F_1 \quad \text{for } i = 1, \dots, q.$$

By (15), $\eta^{-1}\varrho_i \leq r < r_1$, and we can apply (5) to get $W_i \in A(n, m)$ with $e_i \in W_i$ such that

$$(20) \quad A_i = B(e_i, \eta^{-1}\gamma(1-u)\varrho_i/16) \cap W_i \subset F((1-u)\varrho_i/16).$$

Then $b_i \notin P_V A_i$. In fact, if there were $x \in A_i$ with $P_V x = b_i$, we could find by (20) a point $y \in F$ with $|x - y| \leq (1-u)\varrho_i/16$. Then $P_V y \in B(b_i, \varrho_i/2)$ and by (19), (15) and $\eta < \gamma(1-u)/8$,

$$\begin{aligned} d(y, W) &\leq |y - x| + |x - e_i| + d(e_i, W) \\ &< (1-u)\varrho_i/16 + \eta^{-1}\gamma(1-u)\varrho_i/16 + \eta r \\ &< \gamma(1-u)r/4, \end{aligned}$$

whence y would belong to $S_i \cap F$, which is empty.

Let I_i be the closed line segment with end-points b_i and $P_V e_i$. Then $I_i \cap \partial_V P_V(A_i) \neq \emptyset$, since $b_i \in I_i \setminus P_V(A_i)$ and $P_V e_i \in I_i \cap P_V A_i$. Here ∂_V means the boundary relative to V . Since $\partial_V P_V(A_i) = P_V(\partial_{W_i} A_i)$, we can find $a_i \in \partial_{W_i} A_i$ such that $P_V a_i \in I_i$. Let J_i be the closed line segment connecting e_i and a_i . Then $J_i \subset A_i$ and $P_V J_i \subset I_i$, whence by (19)

$$(21) \quad |P_V x - b_i| \leq \varrho_i \quad \text{for } x \in J_i.$$

By (20) J_i is contained in the union of the balls $B(x, \varrho_i)$, $x \in F$. Since the length of J_i is $\eta^{-1}\gamma(1-u)\varrho_i/16$, we can choose a finite number of them, say $B(x_{i,j}, \varrho_i)$, $j = 1, \dots, k$, such that

$$(22) \quad J_i \cap B(x_{i,j}, \varrho_i) \neq \emptyset,$$

$$(23) \quad B(x_{i,j}, \varrho_i) \cap B(x_{i,\ell}, \varrho_i) = \emptyset \quad \text{for } j \neq \ell,$$

$$(24) \quad k > \gamma(1-u)/(160\eta).$$

One can deduce this with the help of Theorem 2.1 or with elementary arguments.

Set

$$B_i = \bigcup_{j=1}^k B(x_{i,j}, \varrho_i) \quad \text{for } i = 1, \dots, q.$$

It follows from (21) and (22) that $P_V B_i \subset B(b_i, 3\varrho_i)$, whence by (13) the sets B_i are disjoint. By (22), (15) and $\eta < (1-u)/8$,

$$|x_{i,j} - e_i| \leq \mathcal{H}^1(J_i) + \varrho_i = \eta^{-1}\gamma(1-u)\varrho_i/16 + \varrho_i < (1-u)r/4.$$

With $b'_i \in B(a, ur) \cap W$, $P_V b'_i = b_i$, as before, (19), (10) and (15) give

$$\begin{aligned} |e_i - b'_i| &\leq |e_i - P_W e_i| + |P_W e_i - b'_i| \leq \eta r + |P_V(P_W e_i) - b_i|/\gamma \\ &\leq \eta r + |P_V(P_W e_i - e_i)|/\gamma + |P_V e_i - b_i|/\gamma \\ &\leq \eta r + \eta r/\gamma + \varrho_i/\gamma \leq 3\eta r/\gamma < (1-u)r/2. \end{aligned}$$

It follows that $B_i \subset B(a, r)$. Using (23), (3) and (24), we obtain

$$\begin{aligned} \mathcal{H}^m(E \cap B_i) &= \sum_{j=1}^k \mathcal{H}^m(E \cap B(x_{i,j}, \varrho_i)) \\ &> k\delta\varrho_i^m > 160^{-1}\gamma(1-u)\eta^{-1}\delta\varrho_i^m \quad \text{for } i = 1, \dots, q, \end{aligned}$$

whence by (8) and (18)

$$\begin{aligned} 3^m r^m &> \mathcal{H}^m(E \cap B(a, r)) \geq \sum_{i=1}^q \mathcal{H}^m(E \cap B_i) \\ &\geq 160^{-1}\gamma(1-u)\eta^{-1}\delta \sum_{i=1}^q \varrho_i^m \\ &> 160^{-1} \cdot 100^{-m}\gamma(1-u)\eta^{-1}\delta tr^m. \end{aligned}$$

But as we may choose for given δ , u and γ an arbitrarily small η , we obtain a contradiction. $\square$

16.3. Remark. Essentially the above proof with $m = 2$, $n = 3$ was given by Marstrand [3] and for general m and n by Mattila [1]. Kirchheim [3] showed that the statement (1) of Theorem 16.2 holds also for m -rectifiable sets in general metric spaces. His proof involves an interesting idea of using semi-norms induced by Lipschitz maps. For a more standard proof of (1) in $\mathbf{R}^n$, see Federer [3, 3.2.19].

Rectifiability and tangent measures

We shall now reformulate the equivalence of rectifiability and weak approximation properties in terms of tangent measures for sets with positive lower density. For this we first give a definition.

16.4. Definition. A measure ν in $\mathbf{R}^n$ is called m -flat if $\nu = c\mathcal{H}^m \llcorner V$ for some $V \in G(n, m)$ and $0 < c < \infty$. The set of m -flat measures in $\mathbf{R}^n$ is denoted by $\mathcal{G}_{n,m}$. For $V \in G(n, m)$ we set

$$\mathcal{G}_{n,m}(V) = \{\nu \in \mathcal{G}_{n,m} : \text{spt } \nu = V\} = \{c\mathcal{H}^m \llcorner V : 0 < c < \infty\}.$$

16.5. Theorem. Let E be an $\mathcal{H}^m$ measurable subset of $\mathbf{R}^n$ with $\mathcal{H}^m(E) < \infty$ and $\Theta_*^m(E, x) > 0$ for $\mathcal{H}^m$ almost all $x \in E$. Then the following are equivalent:

- (1) E is m -rectifiable.
- (2) For $\mathcal{H}^m$ almost all $a \in E$, $\text{Tan}(\mathcal{H}^m \llcorner E, a) = \mathcal{G}_{n,m}(V)$ for some $V \in G(n, m)$.
- (3) For $\mathcal{H}^m$ almost all $a \in E$, $\text{Tan}(\mathcal{H}^m \llcorner E, a) \subset \mathcal{G}_{n,m}$.

Proof. If E is m -rectifiable, then $\Theta^m(E, a) = 1$ and E has a unique approximate tangent plane $V_a = \text{ap Tan}^m(E, a)$ for $\mathcal{H}^m$ almost all $a \in E$ by Theorems 16.2 (1) and 15.19. Using Corollary 14.9 one then finds that for $\mathcal{H}^m$ almost all $a \in E$ every $\nu \in \text{Tan}(\mathcal{H}^m \llcorner E, a)$ is m -uniform with $\text{spt } \nu \subset V_a$. By Theorem 14.11, $\nu = c\mathcal{H}^m \llcorner V_a$, and (2) follows.

Clearly (2) implies (3). Finally, we leave it as an exercise to verify that (3) implies that E is weakly m -linearly approximable and hence m -rectifiable by Theorem 16.2. $\square$

In the next chapter it will be more natural to work with general Radon measures rather than Hausdorff measures on sets. So we now define the rectifiability of measures and state Theorem 16.5 in this context.

16.6. Definition. We say that a Radon measure μ on $\mathbf{R}^n$ is m -rectifiable if $\mu \ll \mathcal{H}^m$ and there exists an m -rectifiable Borel set E such that $\mu(\mathbf{R}^n \setminus E) = 0$.

16.7. Theorem. Suppose that μ is a Radon measure on $\mathbf{R}^n$ with

$$(1) \quad 0 < \Theta_*^m(\mu, x) \leq \Theta^{*m}(\mu, x) < \infty$$

for μ almost all $x \in \mathbf{R}^n$. Then the following conditions are equivalent:

- (2) μ is m -rectifiable.
- (3) For μ almost all $a \in \mathbf{R}^n$, $\text{Tan}(\mu, a) = \mathcal{G}_{n,m}(V)$ for some $V \in \mathcal{G}(n, m)$.
- (4) For μ almost all $a \in \mathbf{R}^n$, $\text{Tan}(\mu, a) \subset \mathcal{G}_{n,m}$.

The density assumptions imply that μ and $\mathcal{H}^m$ are mutually absolutely continuous on the set where (1) holds, recall Theorem 6.9. Hence the theorem follows immediately from Theorem 2.12 (2), Lemma 14.6 and Theorem 16.5.

16.8. Remarks. (1) For sharper results than Theorem 16.7, see Preiss [4, § 5].

(2) It is fairly easy to see that (1) and (2) in Theorem 16.5 are equivalent without the assumption on the positiveness of lower density. However, (3) does not imply them without that assumption. Preiss [4, 5.8–9] constructed an example of a purely 1-unrectifiable Borel set $A \subset \mathbf{R}^2$ with $\mathcal{H}^1(A) < \infty$ for which at $\mathcal{H}^1$ almost all points $a \in A$ every $\nu \in \text{Tan}(\mathcal{H}^1 \llcorner A, a)$ is 1-flat. He also gave there an example of a Borel measure μ on $\mathbf{R}$, singular, with respect to $\mathcal{L}^1$, such that for μ almost all $a \in \mathbf{R}$ every tangent measure of μ at a is a constant multiple of $\mathcal{L}^1$. A different construction with the help of Riesz products for such a measure can be found in Freedman and Pitman [1], see also Kahane [1] and Makarov [2] for singular measures with somewhat similar behaviour.

(3) Theorems 15.19, 16.2, 16.5 and 16.7 give rectifiability criteria in terms of the approximation with m -planes. It is natural to ask if some other classes of sets could be used. From Example 14.2 (3) we see that mere smoothness of the approximating sets is not enough; see also Exercise 4. In the case $m = n - 1$ the proof of Theorem 16.2 can be modified to deal with approximating sets whose complementary components are all convex. This was discovered by David and Semmes in the context of uniform rectifiability, recall 15.23. They developed a Carleson measure condition in this spirit characterizing uniformly $(n - 1)$ -rectifiable sets, see David and Semmes [2]: Theorem I.2.18, Proposition I.2.9 for a related result on m -dimensional sets, and § I.2.2 for a discussion on the general approximation problem.

Lemma 16.1 has also an analogue for uniformly rectifiable sets, see David and Semmes [2, Theorem I.1.76]. Let E be as in 15.23 (1) and recall also the definition of β from 15.23 (4). In the proof of Lemma 16.1 we only used positive lower density and the property (15.9), but not

(15.8), from the definition of the weak approximation property. Thus Lemma 16.1 can be interpreted roughly as saying that if $\beta(x, t)$ is small for small t and if $\mathcal{H}^m(P_V E) > 0$ for some $V \in G(n, m)$, then E cannot be purely unrectifiable. This can easily be turned into a characterization of m -rectifiability. The following is a corresponding result in the context of uniform m -rectifiability.

Let E be as in 15.23 (1). Then E is uniformly m -rectifiable if and only if the following two conditions are satisfied.

(a) There is $\theta > 0$ such that for each $x \in E$ and $r > 0$ there is $V \in G(n, m)$ for which

$$\mathcal{H}^m(P_V(E \cap B(x, r))) \geq \theta r^m.$$

(b) For every $\varepsilon > 0$ the set

$$\{(x, t) \in E \times (0, \infty) : \beta(x, t) > \varepsilon\}$$

is a Carleson set (recall 15.23 (3)).

As was mentioned earlier, in 15.23, condition (b) alone does not imply even m -rectifiability, nor does (a), see e.g. Figure 18.1. The above criterion for uniform rectifiability is somewhat unsatisfactory since it is not given only in terms of projections. Such a projection characterization is still lacking. A natural conjecture is that the validity of (a) for “many” V ’s would imply uniform rectifiability. For the non-uniform rectifiability we shall establish a characterization in terms of projections in Chapter 18.

Exercises.

1. Show that condition (3) of Theorem 16.5 implies that E is weakly m -linearly approximable.
2. Show that conditions (1) and (2) of Theorem 16.5 are equivalent also without the assumption on the positiveness of lower density.
3. Let μ be a Radon measure on $\mathbf{R}^n$ such that $0 < \Theta_*^m(\mu, x) \leq \Theta^{*m}(\mu, x) < \infty$ for μ almost all $x \in \mathbf{R}^n$. Show that if $V \in G(n, m)$ is such that at μ almost all points $a \in \mathbf{R}^n$ there exist tangent measures $\nu \in \text{Tan}(\mu, a)$ with $\text{spt } \nu \subset V$, then $\mathcal{G}_{n,m}(V) \subset \text{Tan}(\mu, a)$ for μ almost all $a \in \mathbf{R}^n$.
4. Give an example of a purely 1-unrectifiable compact set $F \subset \mathbf{R}^2$ such that $0 < \mathcal{H}^1(F) < \infty$, $0 < \Theta_*^1(F, x) \leq \Theta^{*1}(F, x) < \infty$ for $x \in F$ and that for all $a \in F$ the support of every tangent measure $\nu \in \text{Tan}(\mathcal{H}^1 \llcorner F, a)$ is contained in a line. Hint: You may consult David and Semmes [1, § 20].